

Kanishka Singh

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Education

New York University - Courant Institute of Mathematical Sciences	Sept 2025 – Present
M.S. in Computer Science	New York, USA
<i>Relevant Courses: Machine Learning, Vision Meets ML, Predictive Analytics</i>	
Kalinga Institute of Industrial Technology	Sept 2020 – Apr 2024
B.Tech. in Computer Science & Engineering (GPA: 9.04/10.0)	Bhubaneswar, India

Industry Experience

Ernst & Young, Senior Analyst	Jul 2024 – Apr 2025
- Engineered a Stock-Out Prediction Engine using feature correlation analysis and ML models (Logistic Regression, Decision Trees, Gradient Boosting) on 2+ years of inventory data, achieving 85% accuracy on unseen data and enabling data-driven inventory decisions to mitigate stock-out risk. - Built GenAI-powered chatbots for FMCG clients using LangChain, LLM APIs, and retrieval-augmented generation (RAG), delivering CXO-level business insights and significantly reducing decision-making turnaround time. - Resolved ambiguity and contextual understanding challenges in chatbot systems by 90% by implementing LangChain memory modules (ConversationBufferMemory, ConversationSummaryMemory, VectorStoreRetrieverMemory) and designing robust few-shot prompts for improved conversational continuity.	
Ernst & Young, Data Science & Analytics Intern	Jun 2023 – Jul 2024
- Developed “Dashboard Genie”, a GenAI-powered KPI discovery assistant leveraging embeddings, vector search, and LangChain to enable managers to search KPIs across enterprise dashboards, surface top-3 relevant matches with direct links, and significantly reduce time-to-insight and decision-making latency. - Served as Technical Lead for an EV dashboard POC for a major automotive client, re-architecting the backend to handle evolving business logic and implementing ETL pipelines, optimized SQL views, and interactive dashboards to deliver end-to-end analytics. - Designed and automated a Pareto Analysis dashboard using Python and SQL, achieving 80% backend automation and streamlining manual analytical workflows.	

Research Experience

Sentiment-Aware Bitcoin Volatility Forecasting with Residual Decomposition Networks	Sept 2025 – Dec 2025
Prof. Anasse Bari, New York University - Courant Institute of Mathematical Sciences	
- Developed an N-BEATS-based Bitcoin volatility forecasting model with a task-aware hybrid loss, achieving a validation MAE of 0.2897 and outperforming LSTM, CNN, Transformer, and TFT baselines. - Designed a hybrid loss function combining robust error penalization, jump sensitivity, and long-term trend alignment to improve generalization under heavy-tailed, regime-shifting market conditions. - Demonstrated robust performance across volatility regimes, accurately tracking both smooth volatility dynamics and abrupt spike events on held-out test data.	

Projects

Project Scheduling Monitor	— JavaScript, Node.js, MongoDB
- Built a project scheduling and monitoring web application to manage timelines, budgets, and team workflows, implementing backend services and REST APIs within an Agile development environment. - Used JavaScript, Node.js, and MongoDB, ensuring robustness and reliability through unit and integration testing.	

Skills

Languages: Python, JavaScript, TypeScript, Java, C/C++, SQL, R, HTML, CSS
ML/AI Frameworks: PyTorch, TensorFlow, LangChain, Scikit-learn, Matplotlib, PySpark, MATLAB, OpenCV
Technical: Machine Learning, Generative AI, Large Language Models (LLMs), Software Engineering, Web Applications, Backend Development, System Design, REST APIs, Scalable Systems
Methodologies & Tools: Agile Development, Scrum, Kanban, Jira

Achievements

- Awarded “Emerging Extraordinaire” at Ernst & Young for excellence and ownership during the internship.
- Commended by the client’s **Managing Director** with a “**Pat on the Back**” for delivering a **scalable, precise, and user-focused POC** that outperformed a leading consulting firm.